Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME

Pre-Processing

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA
MESTRADO integrado EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA
Inteligência Artificial para as Telecomunicações
2025/2026

Objectives

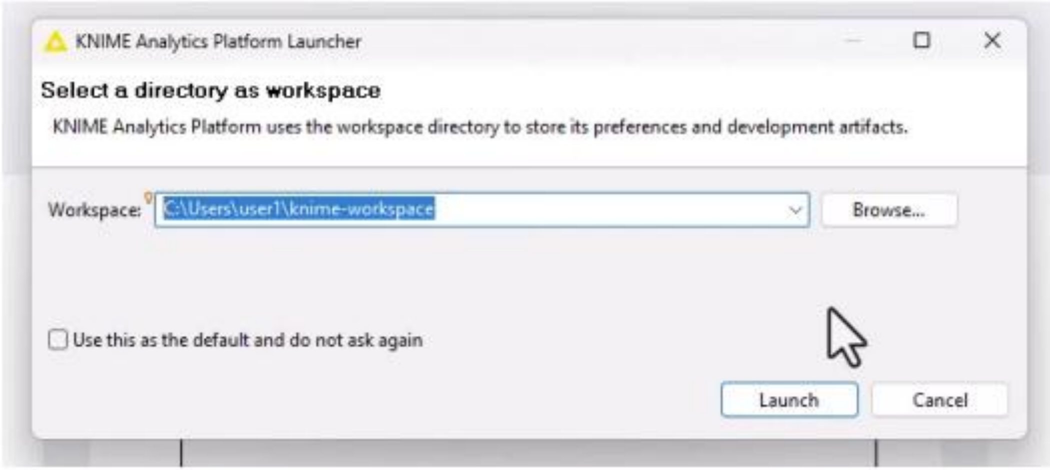
- The primary goal of this task is **to create a comprehensive and efficient workflow** tailored specifically for pre-processing the dataset.

- The pre-processing phase is critical for preparing raw data, ensuring it is clean, well-structured, and suitable for further analysis or modeling.

Pre-processing

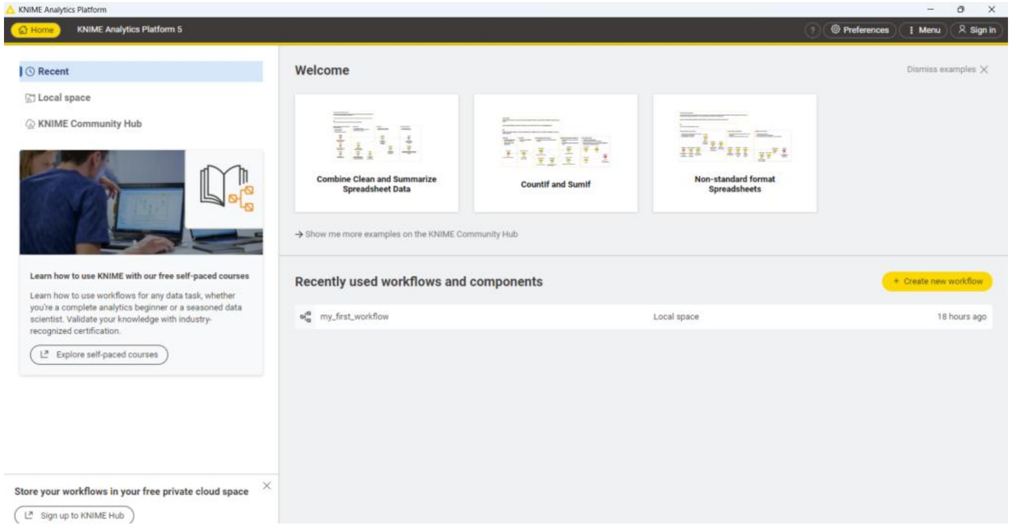
- Data Cleaning: Identifying and handling missing values, correcting inconsistencies, and removing duplicates to ensure data quality.
- Data Transformation: Applying necessary transformations, such as scaling, normalization, and encoding categorical variables, to standardize the dataset.
- Data Integration: Combining or merging multiple datasets to create a unified dataset that is ready for analysis.
- Feature Engineering and Selection: Creating new features, selecting important ones, or reducing dimensionality to improve model performance and computational efficiency.

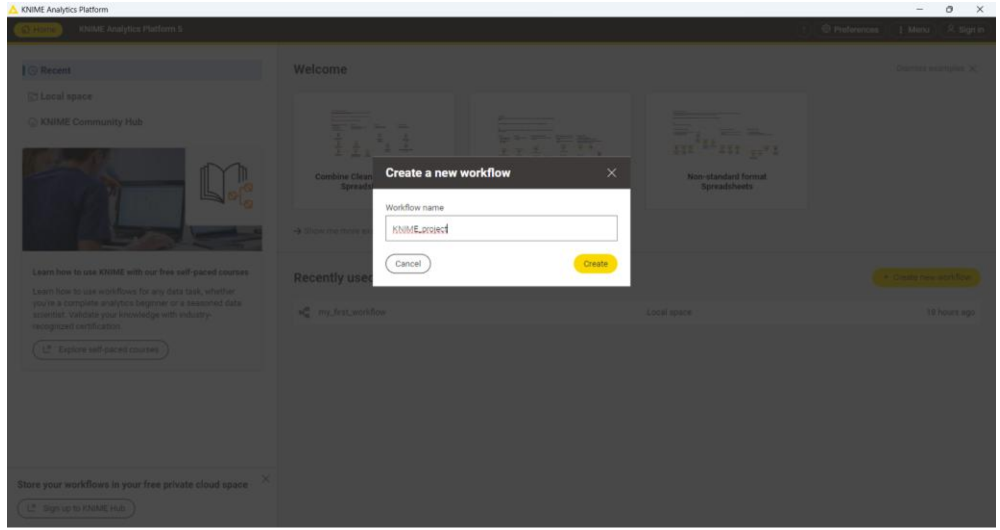
Launching the workspace



- Select a directory as workspace

Creating a new workflow





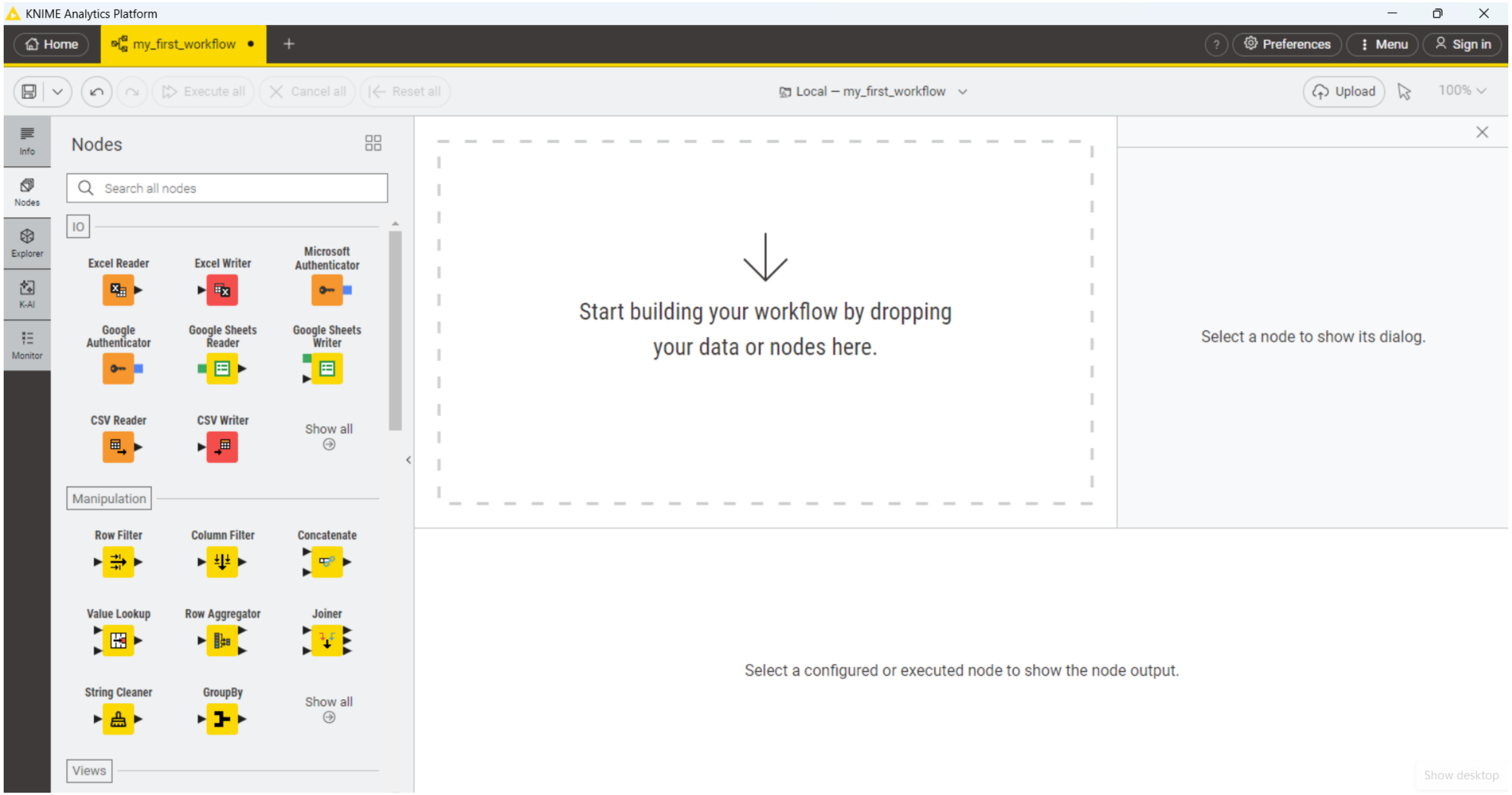
- Click on “Create new workflow”

- Choose a project name
- Click “Create”

ISLab

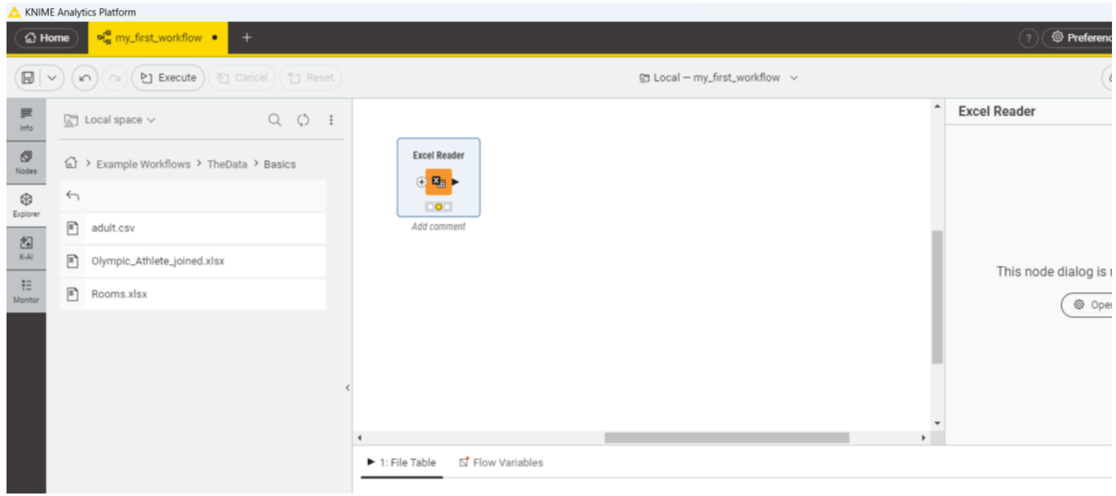
Workflow

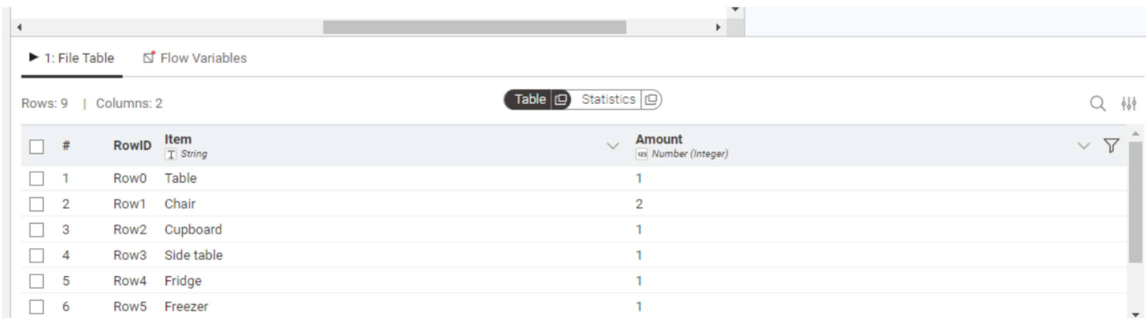
Open for Innovation KNIME



Workflow

- Go to the side panel navigation and open the space explorer tab, under the repository.
- Example Workflows → The Data → Basics.
- Drag and drop the file Rooms.xlsx into the Workflow Editor.
- The node Excel Reader will appear, already configured to read an excel file.
- Execute the Excel Reader node and inspect its output. To do that, move your mouse over the node and click on the play button.
- All the data will appear in the node monitor.
- The excel file provided consists of the amount of certain objects: 1 table, 2 chairs, 1 cupboard, etc.

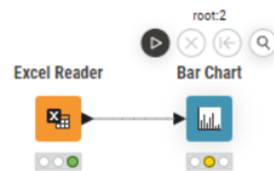


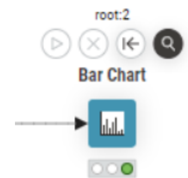


Visualization

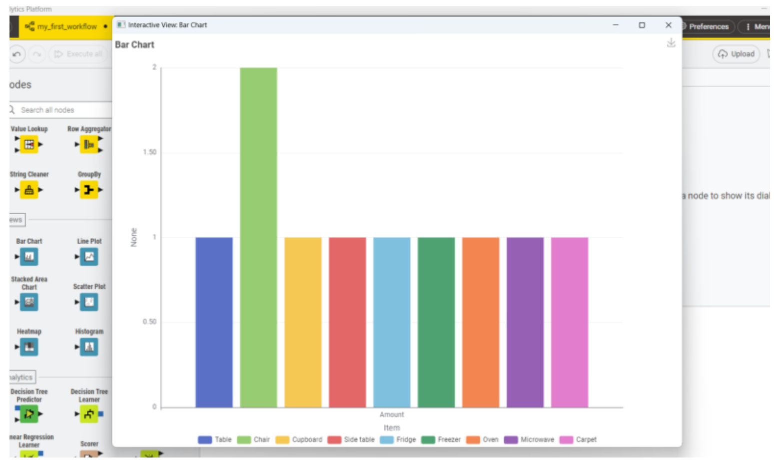
- Go back to the node repository and search for the Bar Chart Node.

- Drag and drop it into the Workflow Editor and connect the output port of the Excel Reader node to the input port of the Bar Chart node.

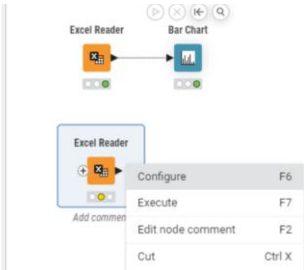




- Hover over the Bar Chart node and click on the magnifier icon to execute the node and open the node view.

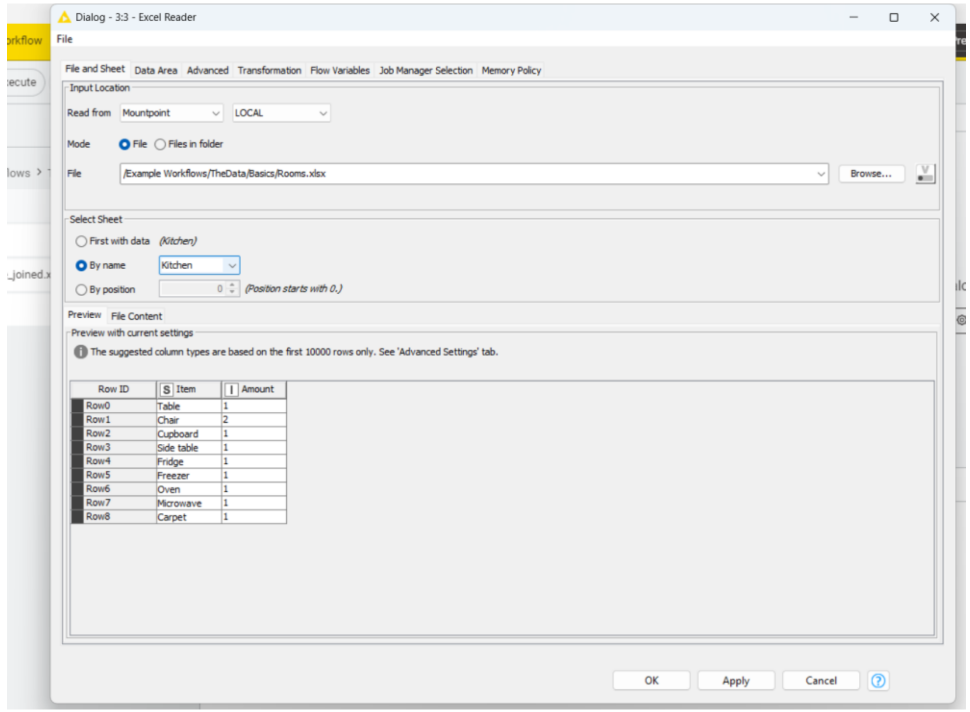


Select Sheet



- Drop the file Rooms.xlsx into the Workflow Editor.
- Open the configuration panel of the node and inspect the settings by right clicking on it and clicking on Configure (or use the F6 shortcut).

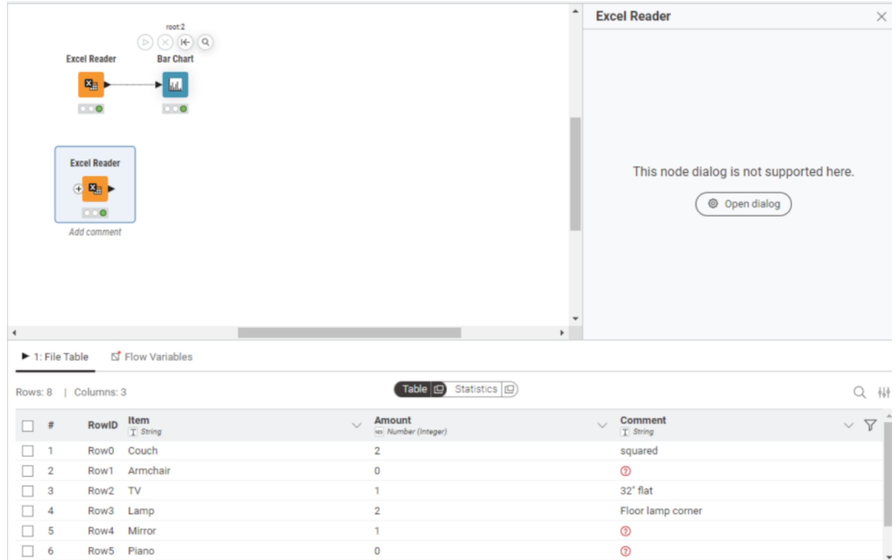
- Here you can see the path to the file you dropped into the workflow editor and a preview of the data table. You can also select the sheet that you want to read the data from.
- First, read the data in the Kitchen sheet. In order to do that, in the Select Sheet section click on By Name and select the Kitchen sheet.
- Execute the Excel reader node by clicking on the play button.
- Now the input data is available at the output port of the Excel Reader node.
- Repeat the exact same process but select the Living Room sheet instead.



Missing Values

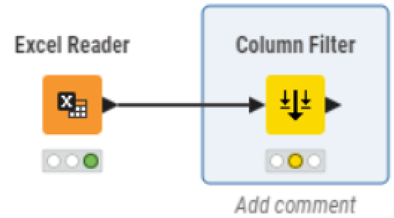
- As you can see in the image above, there are some values consisting of the symbol ?. In KNIME, the ”?” symbol typically represents missing values or unknown values within a dataset.

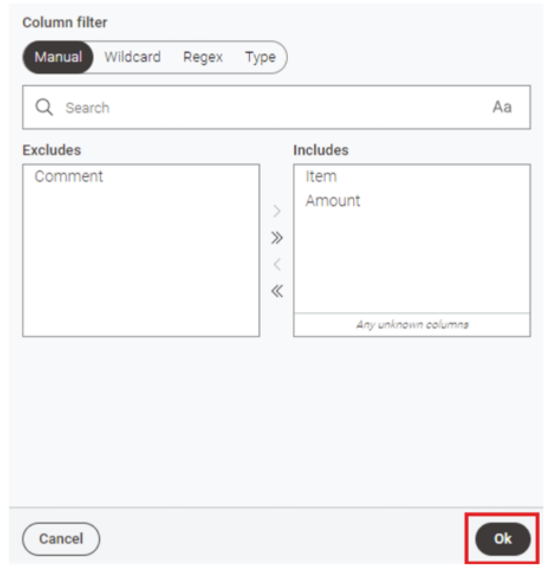
- When you encounter a ”?” value in your data, it indicates that the value for that particular observation or attribute is either missing or unknown. Missing values can arise due to various reasons such as data entry errors, equipment malfunction, or incomplete data collection processes. Handling missing values is an important aspect of data preprocessing and analysis.



Column Filter

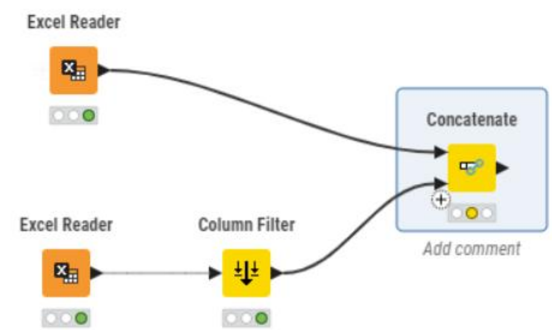
- Search for the Column Filter node in the Node Repository and drag it into the workflow.
- Connect the output port of this second Excel Reader node to the input port of the Column Filter node.
- Open the Column Filter node’s configuration panel. Here, move the column Comment into the Excludes field by first selecting it from the list and then clicking the single arrow pointing to the left.
- Click Apply and Execute.
- Now the filtered data table is available at the output port of the Column Filter node.





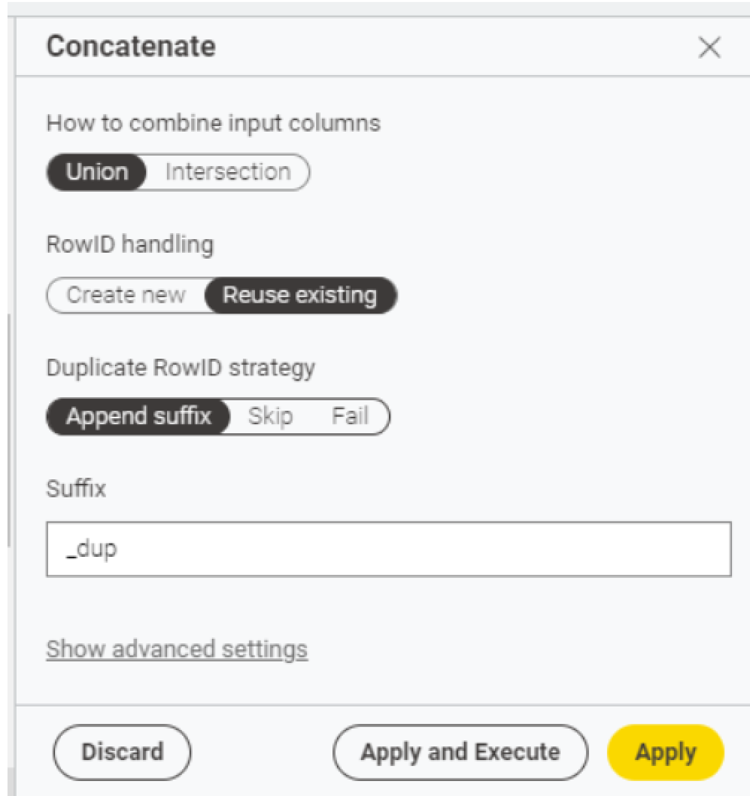
Concatenate

- Now that you have cleaned up the data in the second sheet you can combine all the data in one table.
- The node that enables data combination is the **Concatenate** node.
- Search for it and drag into the Workflow editor. Connect the output port of the Excel Reader node, related to the Kitchen sheet to one of its input ports.
- Connect the output port of the Column Filter node to the last input node.



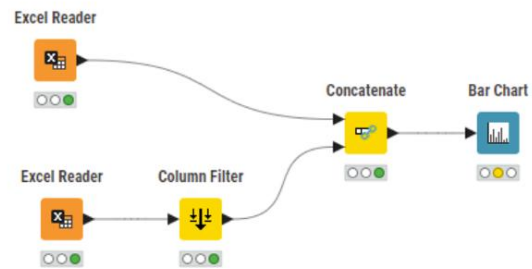
ISLab Concatenate

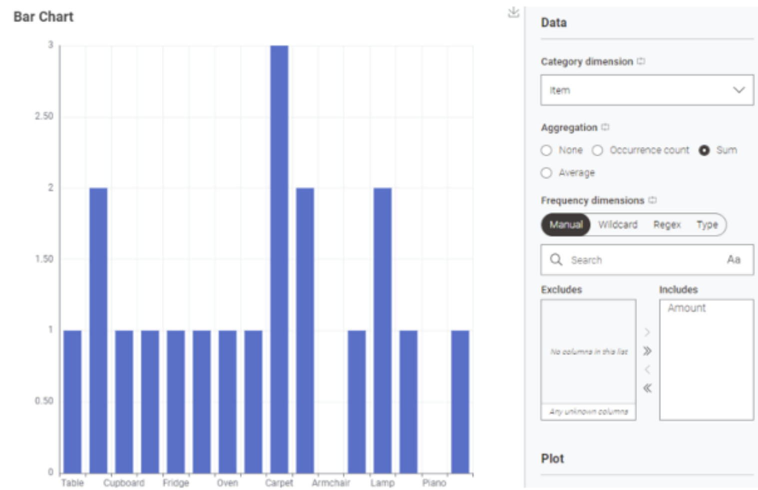
- Now open the Concatenate node configuration panel to select how to combine the input columns and choose what to do in case there are duplicate row identifiers (RowIDs).
- Select Union to use all columns from all input tables and Append suffix so that the output table will contain all rows from both input tables, but duplicate RowIDs are labeled with a suffix.
- Execute the node.



Visualization

- To visualize the data, you will need to use the view nodes.
- Search for the Bar Chart node in the Node Repository. Drag and drop it into the Workflow Editor.
- Connect the output port of the Concatenate node to its input port.
- Open its configuration panel. Here you can configure the node to visualize your data and also have a look at the resulting plot preview right away.
- Select the column that contains the item names as the category dimension;
- Select the sum as the aggregation method; Include the column Amount as the quantitative dimension of the bars.





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